Monitoring Tropical Forest Degradation using Spectral Unmixing and Landsat Time Series Analysis

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Abstract

Tropical forest loss currently contributes 5 to 15% of anthropogenic carbon emissions to the atmosphere. The large uncertainty in emissions estimates is a consequence of many factors, including differences in definitions of forests and degradation, as well as estimation methodologies. However, a primary factor driving uncertainty is an inability to properly account for forest degradation. While remote sensing offers the only practical way of monitoring forest disturbances over large areas, and despite recent improvements in data quality and quantity and processing techniques, remote sensing approaches are still limited in their ability to detect forest degradation. In this paper, a system is presented that uses time series of Landsat data and spectral mixture analysis to detect both degradation and deforestation in forested landscapes. The Landsat data are transformed into spectral endmember fractions and are used to calculate the Normalized Degradation Fraction Index (NDFI; Souza et al., 2005). The spectrally unmixed data are used for disturbance monitoring and land cover classification via time series analysis. To assess the performance of the system, maps of deforestation and degradation were used to stratify the study area for collection of sample data to which unbiased estimators were applied to produce accuracy and area estimates of degradation and deforestation from 1990 to 2013. The approach extends previous research in spectral mixture analysis for identifying forest degradation to the temporal domain. The method was applied using the Google Earth Engine and tested in the Brazilian State of Rondônia. Degradation and deforestation were mapped with 88.0% and 93.3% User's Accuracy, and 68.1% and 85.3% Producer's Accuracy. Area estimates

of degradation and deforestation were produced with margins of error of 13.9% and 5.3%, respectively, over the 24 year time period. These results indicate that for Rondônia a decreasing trend in deforestation after 2004 corresponds to an increase in degradation during the same time period.

Introduction

While there is no internationally accepted definition of forest degradation, it can generally be defined as a disturbance caused by human intervention in a forested landscape that results in carbon emissions but not a change in land cover (GFOI, 2016, p. 19). Compared to intact forests, degraded forests are often lower in biomass, less ecologically productive, fragmented, and lower in canopy cover (Edwards et al., 2011). However, information on the rates and locations of degradation vary widely; global estimates of the carbon emissions from degradation range from 40 to 212% of those for deforestation (Baccini et al., 2017; Berenguer et al., 2014; Pearson et al., 2017). This large uncertainty is primarily due to the difficulties in monitoring degradation over large areas and, as a consequence, large uncertainties remain in the role of forest disturbances in the terrestrial carbon cycle (Pan et al., 2011).

Developing countries are eligible for monetary compensation for emissions reductions associated with reduced rates of degradation through the United Nation's Reduced Emissions from Deforestation and Degradation (REDD+) Program (UNFCCC, 2014). In fact, forest degradation is identified as one of the key REDD+ activities in the UNFCCC Cancun Agreements (United Nations, 2010). The IPCC

recommends the use of activity data (i.e. information on anthropogenic activities that cause emissions or removals of atmospheric carbon, such as deforestation and degradation), and emission factors (changes in carbon stock caused by these activities), for estimation of emissions and removals of greenhouse gases over large areas (Milne and Jallow, 2005). The activity data are typically estimated from sample data that is obtained under sampling designs in which change maps are used to stratify the study area (GFOI, 2016). The most practical way of creating maps of change over large areas is the use of remote sensing data. Remote sensing data are spatially explicit, provide an historical context, and are available at medium-to-high spatial resolution, often at no cost to the user (Herold et al., 2011; Woodcock et al., 2008). Remote sensing approaches for monitoring forest cover change have been successfully implemented for global carbon accounting (Hansen et al., 2013; Harris et al., 2012) and in support of REDD+ reporting systems (Goetz et al., 2014). However, while forest degradation is a REDD+ activity, no available methodology has proven capable of routinely mapping degradation (Herold et al., 2011). Therefore, most REDD+ monitoring systems do not actively quantify and report on emissions related to degradation (Hosonuma et al., 2012).

Rates and areas of land change can be mapped from remote sensing data using image-processing techniques that are based on change detection or classification. Change detection is possible because changes in land surface conditions alter the spectral signature measured by the satellite, but the subtle and spatially-isolated nature of degradation makes remote sensing of forest degradation difficult (Goetz et al., 2009). Furthermore, degradation events often occur at a smaller spatial scale

than easily available remote sensing data. High-resolution data has been used in the past (Asner et al., 2013; Franke et al., 2012; Rahm et al., 2013), but the cost of acquiring multiple images over the same area makes routine monitoring problematic (Goetz et al., 2014). This limitation has meant that many approaches to monitoring degradation using high resolution imagery have utilized only one observation for the study region (Aalders et al., 2015; Asner et al., 2013; Connette et al., 2016).

There are significant limitations to approaches for mapping degradation that rely on a single image. Clouds, cloud shadows, atmospheric constituents, or missing or mis-registered data can cause errors or exclude large regions from analysis. The use of single-date images for detecting degradation has often relied on identifying canopy gaps (Asner et al., 2013), which is problematic as many tropical environments naturally contain non-continuous canopies due to climatic or geologic limitations (Hirsch, 1990). For example, an estimated 22% of Brazil consists of open-canopy Cerrado forests, second only to the Amazon rainforest in size among major biomes (Marquis, 2002; Ratter et al., 1997). Therefore, an approach to monitoring degradation that is based on canopy cover alone is not a robust indicator of degradation in open canopy forests.

For accurate monitoring of degradation events, historical imagery and frequent image acquisitions are important (Herold et al., 2011), but the use of such data is not possible if relying on high resolution data. For small-scale events such as selective logging, the regrowth of vegetation can occur quickly after the event, causing the resulting canopy damage to be indistinguishable from the surrounding forests in a

short amount of time (Souza and Roberts, 2005). Alternative causes of degradation, such as biomass removal for fuelwood collection, can be a gradual process. Thus, the use of high-resolution imagery prohibits the characterization of both the history and the post-disturbance dynamics of the landscape, which is required for estimation of the carbon emissions associated with the degradation event. The temporal variations and the importance of the landscape history imply that continuous monitoring and minimal gaps between image acquisitions is preferable (Lambin, 1999). These issues suggest that a robust system for monitoring degradation would take into account both the spatial and temporal attributes of the disturbances, and thus should rely primarily on repeated observations from a system like Landsat or Sentinel-2.

Unlike high-resolution imagery, Landsat data are freely available at 8-to-16 day intervals at 30-meter spatial resolution. The free data policy of the Landsat program combined with a relatively high observation frequency and a spatial resolution suitable for environmental remote sensing have allowed the development of land monitoring algorithms that take advantage of the seasonal and/or long-term trends in the data (Brooks et al., 2014; Kennedy et al., 2010; Verbesselt et al., 2010; Zhu and Woodcock, 2014). Many of these techniques utilize classical approaches to structural break detection for detecting disturbances in a time series. Time series-based approaches are less susceptible to erroneous results due to image "noise" and are capable of characterizing both long-term and abrupt changes to landscapes (Zhu, 2017).

To date, time series-based approaches have usually been used to monitor disturbance events that are larger than the spatial resolution of the data. Degradation events, however, frequently occur at a scale smaller than 30 meters (Herold et al., 2011). An approach to monitoring sub-pixel degradation events with individual Landsat images has been the use of spectral mixture analysis (Souza and Barreto, 2000). Spectral mixture analysis assumes that every pixel is composed of varying proportions of spectrally pure endmembers (Adams et al., 1986). The proportion of each endmember is dependent on the composition and structure of the physical elements in the pixel. In this approach, a forested pixel would not need to be entirely cleared to alter the sub-pixel endmember proportions. Souza et al. (2013) utilized spectral mixture analysis for monitoring degradation in the Brazilian Amazon. However, the approach utilized a single image for each year and was therefore susceptible to the issues outlined above. Additionally, their methodology was limited to monitoring degradation in dense canopies and excluded open-canopy forests.

The goal of this study was to develop a new methodology that is capable of mapping and estimating degradation and deforestation over large areas. Specifically, the goals were to: 1) extend the methodology utilized in Souza et al. (2013) to the temporal domain for continuous monitoring of forest degradation and deforestation; and 2) apply the methodology to the Brazilian State of Rondônia and estimate areas of deforested and degraded land from 1990 to 2013.

Methodology and Study Area

Study area



Figure 1. The study area located in the Brazilian State of Rondônia (red grid).

Rondônia is located in Western Brazil along the border with Bolivia (Figure 1). The Amazon tropical forest biome covers the majority of Rondônia, while large swaths of Cerrado occur in the southern portion of the State (Eiten, 1972). Deforestation in Rondônia was minimal until the 1960s (Brondizio and Moran, 2012). The development of the BR-364 highway in the 1960s provided a link between Rondônia and the Atlantic ocean, eventually leading to large-scale deforestation due to expansion of cattle pastures and agriculture (Barona et al., 2010; Browder and Godfrey, 1997). Small-scale clearings for logging, shifting cultivation, or ranching have also been an influential driver in forest conversion (Pedlowski et al., 1997). Much of the logging has occurred illegally in dedicated conservation areas

(Chomentowski et al., 2005). Natural events, such as fire, windfall, and flooding are common and can also disturb forests. According to SEDAM, Rondônia's Secretary of Environmental Development, 24% of Rondônia's original forested land had been converted to other land covers by 1996 (SEDAM, 1996).

Like the rest of Brazil, deforestation in Rondônia began to slow after 2004 due to the implementation of the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm) (PPCDAm, 2004). PPCDAm consisted of a series of multi-sectoral policies aimed at incentivizing conservation and de-incentivizing forest conversion. While Brazil's National Institute for Space Research (INPE) has tracked deforestation since the 1970s, no such information is currently available for degradation (INPE, 2000). Therefore, the effects of the recent policies on the rates of degradation in the area are not well understood. To explore long-term trends in degradation we defined our study period as being from 1990 to 2013.

Data

All available Landsat Collection 1 surface reflectance data were utilized for the analysis. Collection 1 data is converted to surface reflectance using the LEDAPS (TM/ETM+) and LaSRC (OLI) algorithms (Masek et al., 2012; Vermote et al., 2016). The pre-processing and disturbance monitoring was performed using the Google Earth Engine (GEE) Python and Javascript APIs (Gorelick et al., 2017). Training data for land cover classification were collected on the GEE browser interface.

190 *Deforestation and Degradation Definitions*

191 The United Nations Food and Agriculture Organization's (FAO) Forest Research
192 Assessment (FAO, 2015, Terms and Definitions) defines deforestation as "The
193 conversion of forest to other land use or the permanent reduction of the tree canopy
194 cover below the minimum 10 percent threshold". Additionally, the definition
195 "excludes areas where the trees have been removed as a result of harvesting or
196 logging, and where the forest is expected to regenerate naturally or with the aid of
197 silvicultural measures". FAO's definition of forest degradation, however, is less
198 directly relatable to information that is detectable with remote sensing imagery or
199 relevant to carbon inventories. Degradation is defined as the "reduction of the
200 capacity of a forest to provide goods and services" (FAO, 2015, Terms and
201 Definitions).

202 In the IPCC Good Practice Guidelines (IPCC, 2006; Penman et al., 2003) forest
203 degradation is not identified by name but the guidelines imply it "can be estimated
204 as the effect on emissions and removals of human interventions on land continuing
205 to be used as forests" (GFOI, 2016, p. 19), and further that "degradation is
206 interpreted here as the processes leading to long-term loss of carbon without land-
207 use change, otherwise there would be deforestation" (GFOI, 2016, p. 61).
208 Accordingly, no clear formal definition of forest degradation exists. In this paper, we
209 define forest degradation as: *a loss of carbon from forestland without a change in*
210 *land cover caused by either natural or anthropogenic activities*. According to this
211 definition, degradation can include, but is not limited to, selective logging, fires,
212 windfall, logging that is immediately followed by regrowth or forest management,

213 damage from surrounding activities, flooding, and insect damage. While it is not
214 implicitly stated how we define degradation and deforestation, it is likely that both
215 will alter the composition and/or structure of the forest (Edwards et al., 2011).
216 Accordingly, we refer to them both as *forest disturbances* and a specific occurrence
217 as a *disturbance event.*

218 *Spectral Mixture Analysis and NDFI*

219 Souza et al. (2005) related field data on cleared, selectively logged, and non-
220 disturbed forests in the Amazon to proportions of Soil, Shade, green vegetation (GV),
221 and non-photosynthetic vegetation (NPV) using a linear mixture model. Dense
222 forests contained large fractions of GV, low Soil and NPV fractions, and lower Shade
223 than the other land covers in the area. Cleared forests contained high Soil and NPV
224 fractions and thinned forests contained higher canopy Shade and lower GV than
225 non-disturbed forests. A ratio of endmember fractions, the Normalized Degradation
226 Fraction Index (NDFI), was developed to emphasize the difference between non-
227 disturbed, thinned, and cleared forest pixels. NDFI is defined as:
228

$$NDFI = \frac{GV_{Shade} - (NPV + Soil)}{GV_{Shade} + (NPV + Soil)}$$

229 where,

$$GV_{shade} = \frac{GV}{1 - Shade}$$

230

231 NDFI values range from -1 to 1. Dense canopy forests have an NDFI value close
232 to 1. A decrease in tree cover will result in a decrease in NDFI and a complete

clearing of the forest results in a value approaching -1. Open canopy forests have NDFI values similar to dense forests that have been thinned due to their similar canopy structure (Figure 2).

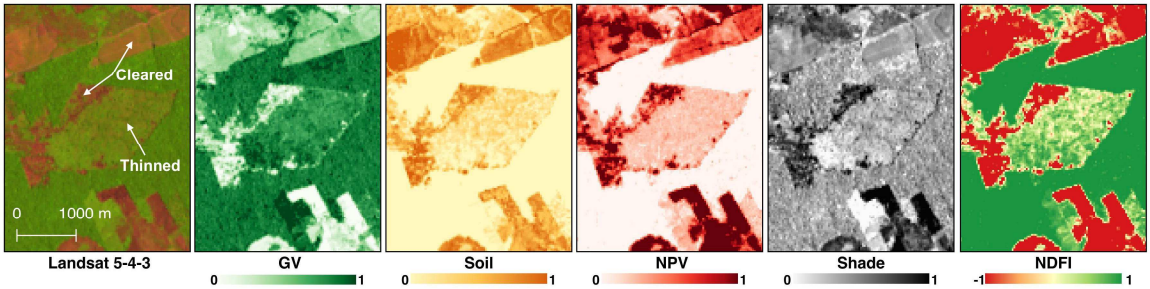


Figure 2. An example of a logged forest in an image that has been transformed into proportion GV, Soil, NPV, and Shade, and used to calculate NDFI. Notice that the locations that have been cleared have lower proportion GV, higher proportion Soil, NPV, and Shade, and higher NDFI than areas that have been only thinned.

Change detection and classification algorithm

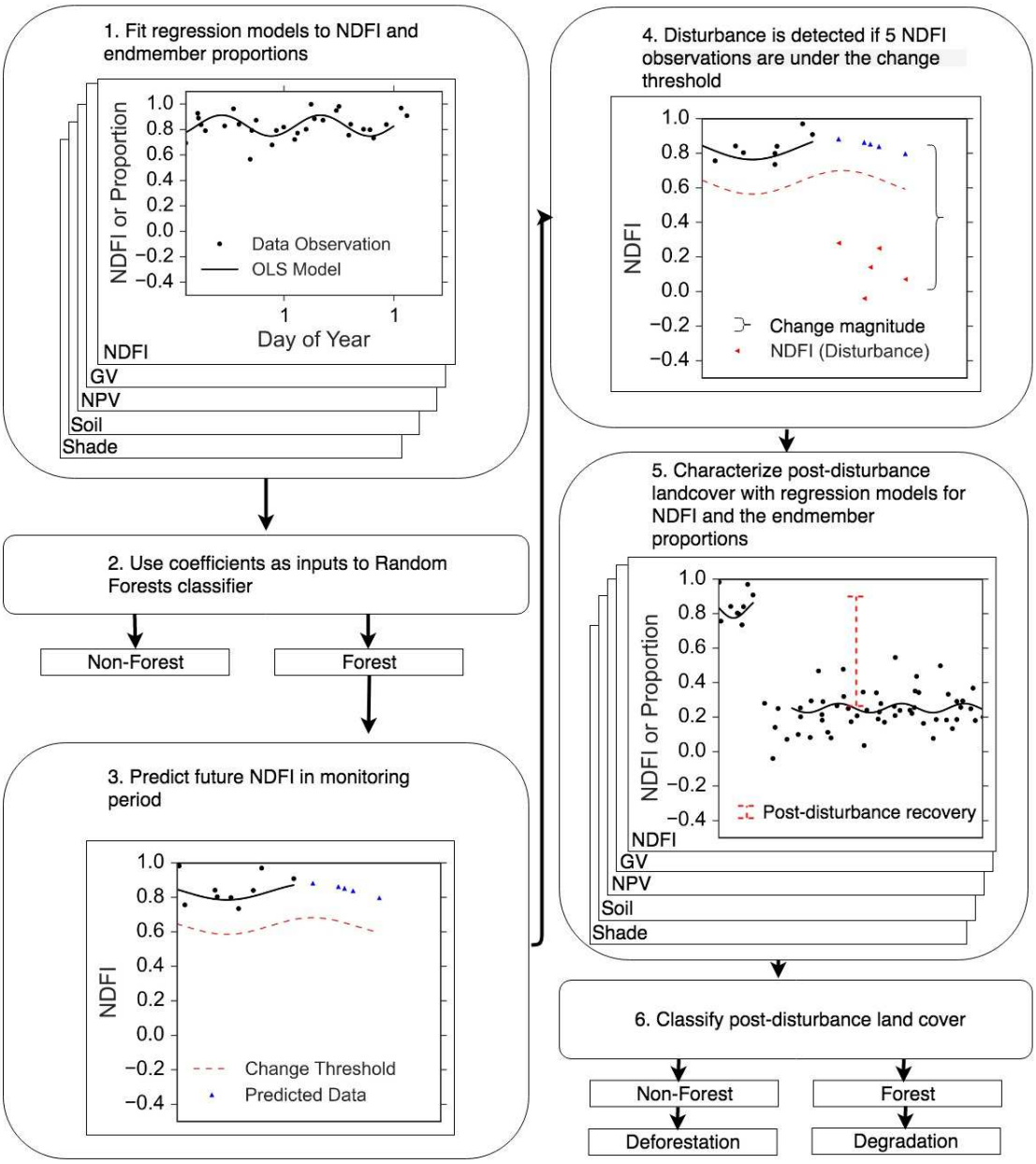


Figure 3. Schematic representation of the disturbance monitoring process.

All Landsat images used in the analysis were transformed into endmember fractions representing proportion GV, NPV, Soil, Shade, and Cloud using the 'unmix' function in GEE. The unmixing algorithm is a simple linear mixture model that uses a single set of endmembers for unmixing each image. We constrained the model to return

endmember proportions that sum to one and are non-zero. The spectral endmembers in Souza et al. (2005) were used in this analysis, with the exception of the cloud endmember (Table 1). A total of 136 images from seven WRS-2 scenes in Rondônia were used to calculate candidate cloud endmembers using the N-FINDR algorithm (Winter, 1999). The cloud endmember was used to exclude cloud-contaminated pixels that were missed by FMask. The endmember fractions are then used to calculate NDFI.

Table 1. Spectral endmembers used for spectral mixture analysis. The endmembers were used to estimate the fractions of each endmember in each pixel. The GV, NPV, Soil, and Shade endmembers were adopted from (Souza et al., 2005) while the Cloud endmember was derived from 136 potential endmembers from images in the study region.

Endmember	B1	B2	B3	B4	B5	B7
GV	0.05	0.09	0.04	0.61	0.30	0.10
NPV	0.14	0.17	0.22	0.30	0.55	0.30
Soil	0.20	0.30	0.34	0.58	0.60	0.58
Shade	0.00	0.00	0.00	0.00	0.00	0.00
Cloud	0.90	0.96	0.80	0.78	0.72	0.65

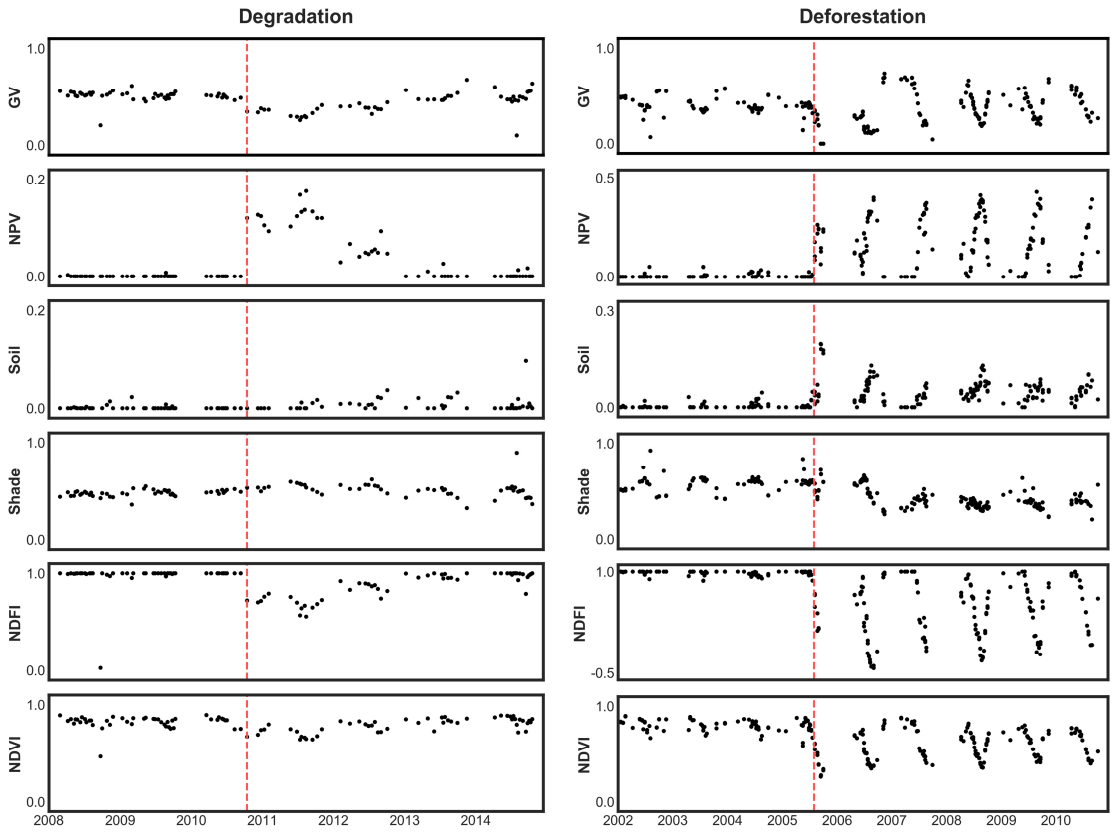


Figure 4. The time series for two pixels that undergo degradation (left column) and deforestation (right column). The date of disturbance is indicated with the red dashed line. The data represent all cloud-free observations for the pixels for GV, NPV, Soil, Shade, and NDFI. NDVI is shown for comparison. Notice that NDVI has more variability during the period of stable forest, and that the disturbances result in a lower magnitude change than in NDFI.

The change detection algorithm operates on the time series of each pixel in the study region (Figure 4). Ordinary least square (OLS) regression models are fit to the time series of all endmember fractions and NDFI data.

$$y_{(i,x)} = \beta_0 + \beta_1 x + \beta_2 cos\left(\frac{2\pi}{365.25}x\right) + \beta_3 sin\left(\frac{2\pi}{365.25}x\right)$$

where,

x = day of year

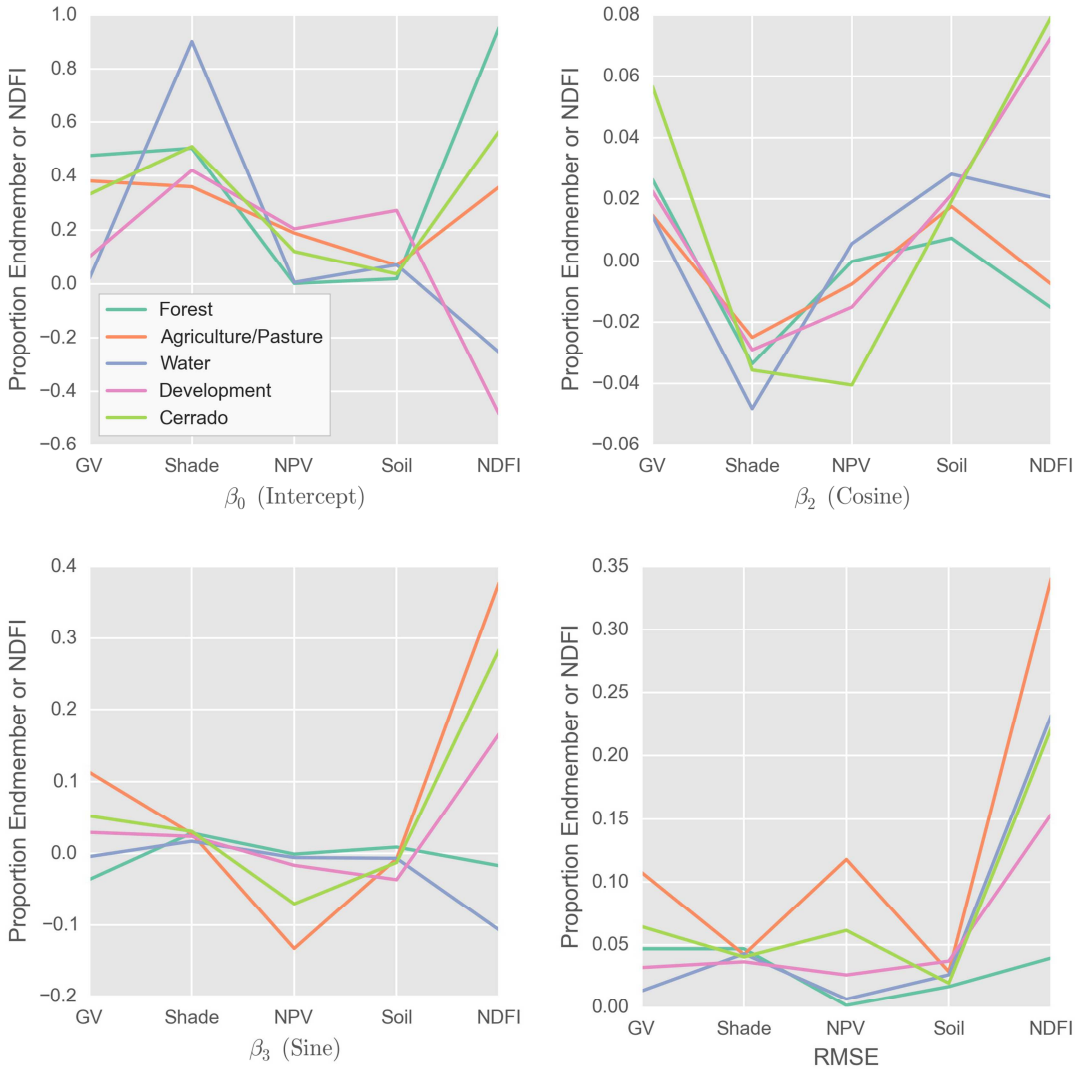
$y_{(i,x)}$ = predicted endmember or NDFI i at time x

β_0 = coefficient for overall magnitude or intercept

β_1 = coefficient for inter-annual variability or slope

$\beta_{2,3}$ = coefficients for intra-annual variability

The OLS models were fit for an initial training period of 1984-1989. The intercept (normalized to the mid-point of the model period) and intra-annual variability coefficients in addition to the model root-mean-squared error (RMSE) were used as inputs to a Random Forests classifier to assign land cover labels for the training period. A total of 1,250 training pixels were selected for the land covers of *forest*, *agriculture or pasture*, *Cerrado* or *natural grassland, developed*, and *water* (Figure 6). Locations of *Cerrado* that contained over 10% tree cover were included in the *forest* class. Only locations classified as *forest* for the training period were used in the disturbance monitoring.



290

291 **Figure 5. Average values for the 20 inputs used for land cover classification derived**
292 **from the training data from 1984-1989. The regression coefficients and RMSE for the**
293 **four endmembers and NDFI are used to assign land cover labels and every location**
294 **that are classified as *forests* are used in the subsequent change detection process.**
295 **Notice that the *forests* class has high intercepts for NDFI and GV and low RMSE,**
296 ***agriculture* tends to have high RMSE for most proportions, *water* has a high Shade**
297 **intercept, *development* has a low NDFI intercept, and *Cerrado* has large cosine terms**
298 **for GV and NDFI. These differences help separate the classes during the land cover**
299 **classification stage of the algorithm.**

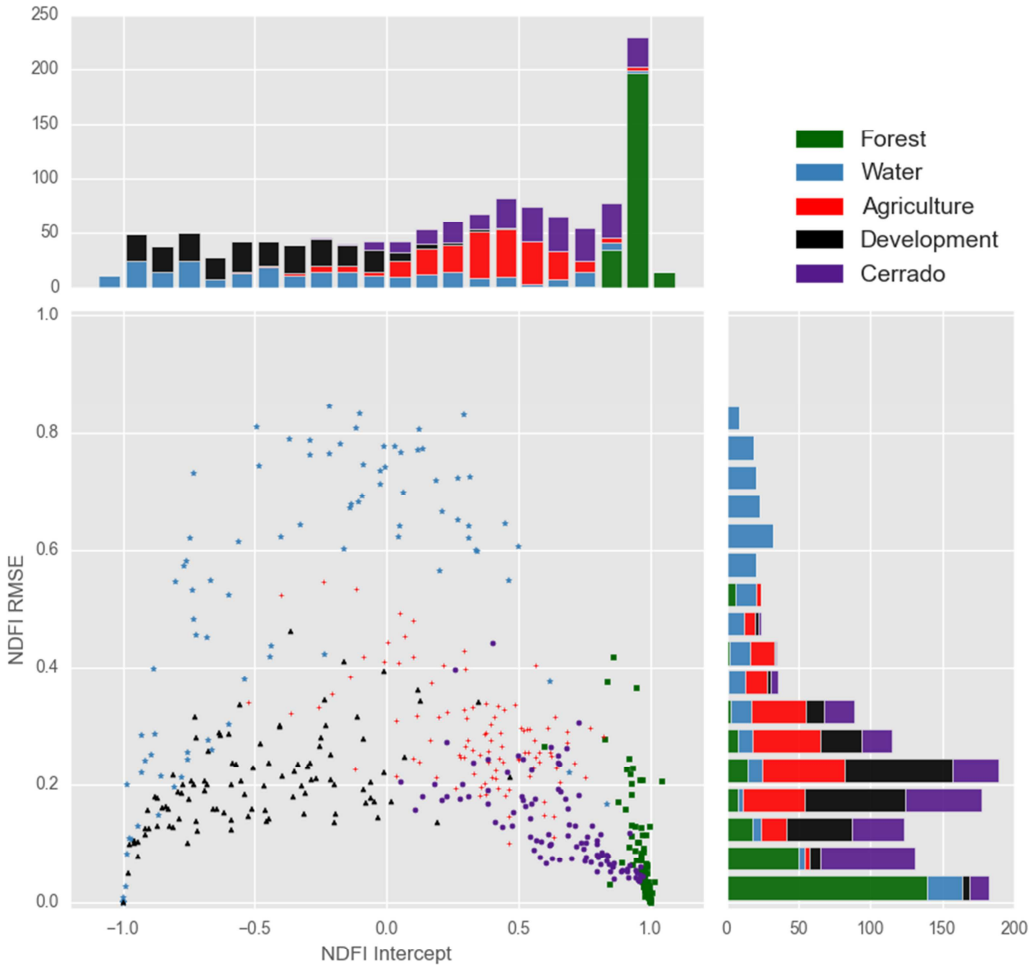
300

301 The disturbance monitoring was performed using NDFI during the period 1990-

302 2013. Previous research has shown NDFI to be more sensitive to tropical forest

303 disturbances than other spectral indices (Schultz et al., 2016). However, we found

304 that water and agriculture can occasionally have NDFI values similar to forests
305 (Figure 6). Therefore, we found it sufficient to rely on NDFI for change detection but
306 include the regression coefficients for GV, NPV, Soil, and Shade as inputs for land
307 cover classification.



308

309 **Figure 6. Scatterplot of two of the inputs to land cover classification from the**
310 **regression models. Points represent training sample locations during the training**
311 **period. These inputs are just 2 of 20 input variables used for land cover classification**
312 **(see Figure 5).**

313

The disturbance monitoring algorithm is closely related to the Continuous Change Detection and Classification algorithm (CCDC; Zhu and Woodcock, 2014) but altered to target degradation and deforestation. The OLS model during the training period was used to predict future NDFI observations. The residuals of the observations were normalized by the training model RMSE to calculate a *change score* for each observation, which is roughly equivalent to a Z-score. If the change score was below a threshold of -5 for five consecutive observations then the model indicated a disturbance (Figure 3). The motivation to use such a high value for the change score and consecutive observations is that deviations beyond the average variability of the data are common due to a variety of causes ranging from missed clouds to noise in the imagery or uncertainties in atmospheric correction process as well as minor changes in vegetation state or condition. The change score is dependent on the model RMSE and, consequently, a pixel with low natural variability during the training period will be sensitive to more subtle disturbances than one with high variability. The RMSE of NDFI for a stable forest is generally low, suggesting that it would be sensitive to low magnitude disturbances (see Figures 4, 5). While this indicates that NDFI would also be sensitive to ephemeral "noise" in the image data, the requirement that multiple consecutive observations exceed the change threshold minimizes the chance of false change detection.

An example of a time series of NDFI for a pixel that undergoes a degradation event can be seen in Figure 7. The location of the event is in a heavily fragmented area, much of which has been converted to pasture or agriculture. Around 2009 a fire or thinning event caused a decrease in NDFI which triggered the detection of the

disturbance. The decrease in NDFI caused by the disturbance, quantified as the average change score over the five-observation window, is recorded as the *change magnitude*. Lower NDFI values correspond to higher proportions of Soil or NPV. Therefore, the change magnitude is assumed to be an approximate indicator of the amount of tree removal or canopy damage that occurred due to the disturbance. The change magnitude was later used in the post-processing stage when identifying possible false disturbance labels.

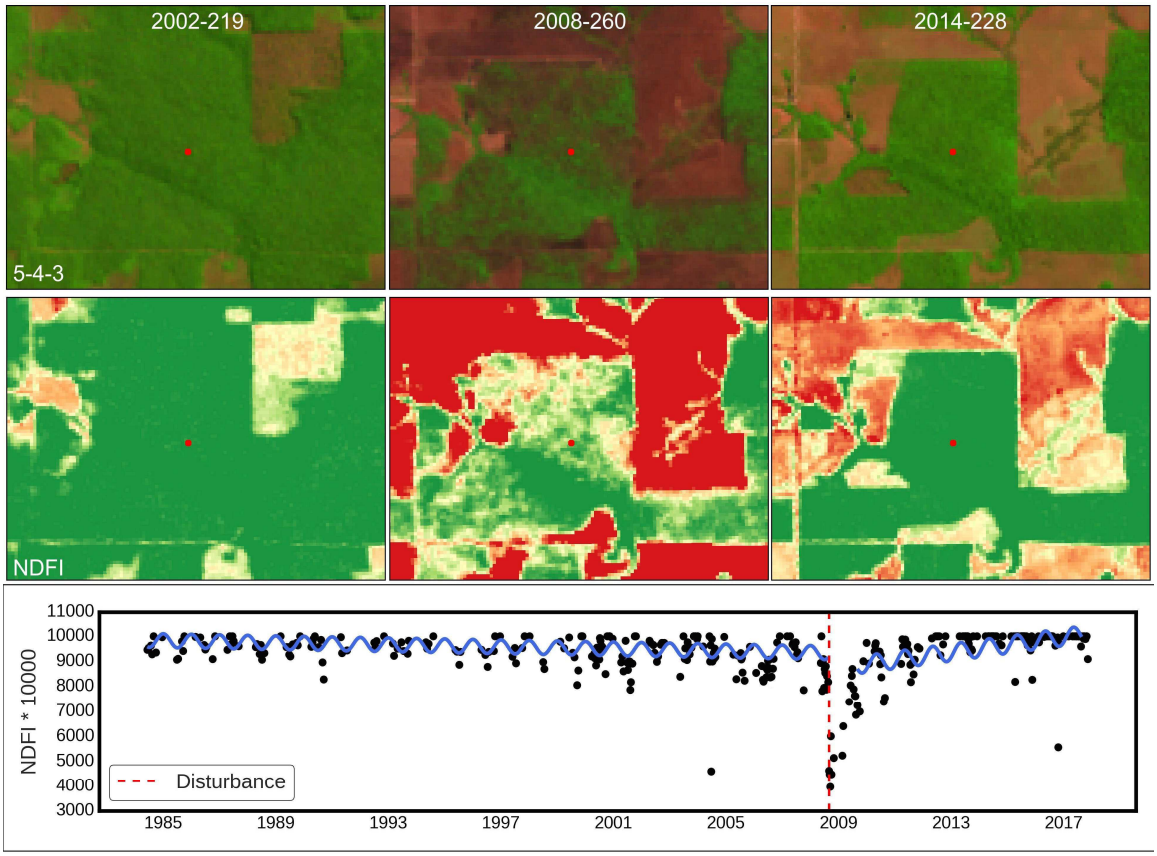


Figure 7. An example of a pixel that undergoes a degradation event. The top row contains Landsat 5-4-3 composites before (left), during (center) and after (right) the disturbance event. The middle row contains the same images transformed into NDFI. The bottom is the NDFI time series for the pixel labeled in red in the images. Since the disturbance did not result in a land cover change it is labeled as a degradation event.

After detecting the disturbance, a new regression model is fit to the NDFI data and endmember fractions. The models after the disturbance begin a year after the event is detected to avoid fitting the model to the early stages of forest regrowth in degradation events (Figure 7). The ratio of the intercept term for the NDFI regression model before the disturbance to the intercept term after the disturbance is recorded as the *post-disturbance recovery* (Figure 3*)*. In Figure 7, the recovery is fast and within five years the NDFI returns to a similar magnitude as it was before the disturbance. The post-disturbance recovery is used later to identify degradation events that were mis-classified as deforestation. The regression coefficients in the model after the disturbance are used for post-disturbance land cover classification. A Random Forests classifier is used to assign land cover labels to every time segment (Figure 5) such that a land cover label is associated with every location and time period within the study area. Disturbances that result in a change in land cover are labeled as *deforestation*, and those that do not as *degradation*. As can be seen in the Landsat imagery in Figure 7, the location does not undergo a landcover change and is therefore classified as *degradation*.

Based on the training data for stable forests during the training period, the change threshold was exceeded in just 5% of the observations, which indicates that in 5% of the pixels there are factors such as clouds or smoke that incorrectly produce the conditions that would indicate a change. Since these factors are ephemeral, however, the five consecutive observation requirement prevents an incorrect classification of disturbance (Figure 8). The requirement of consecutive

observations reduces errors of commission, but may also result in errors of omission if the disturbance event is only apparent for less than five observations.

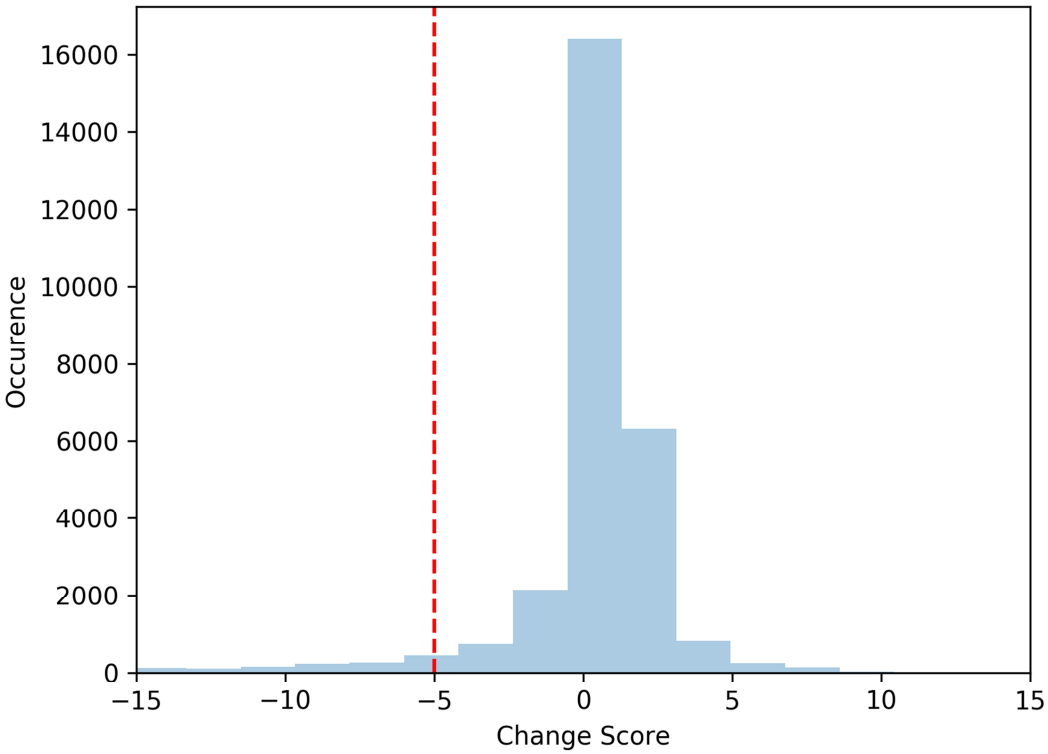


Figure 8. Change scores for all images in the training period for the training data. The change threshold in indicated in the red dotted line. If 5 consecutive observations have a change score lower than the threshold a disturbance is identified.

Since the model runs continuously through time, multiple degradation or deforestation events can be detected during the study period. Figure 9 shows an example of a location that undergoes two disturbances. The first disturbance does not result in complete removal of the canopy, and following the disturbance the forest appears to regrow. The second disturbance results in a land cover change. Therefore, the first disturbance label is degradation and the second is deforestation.

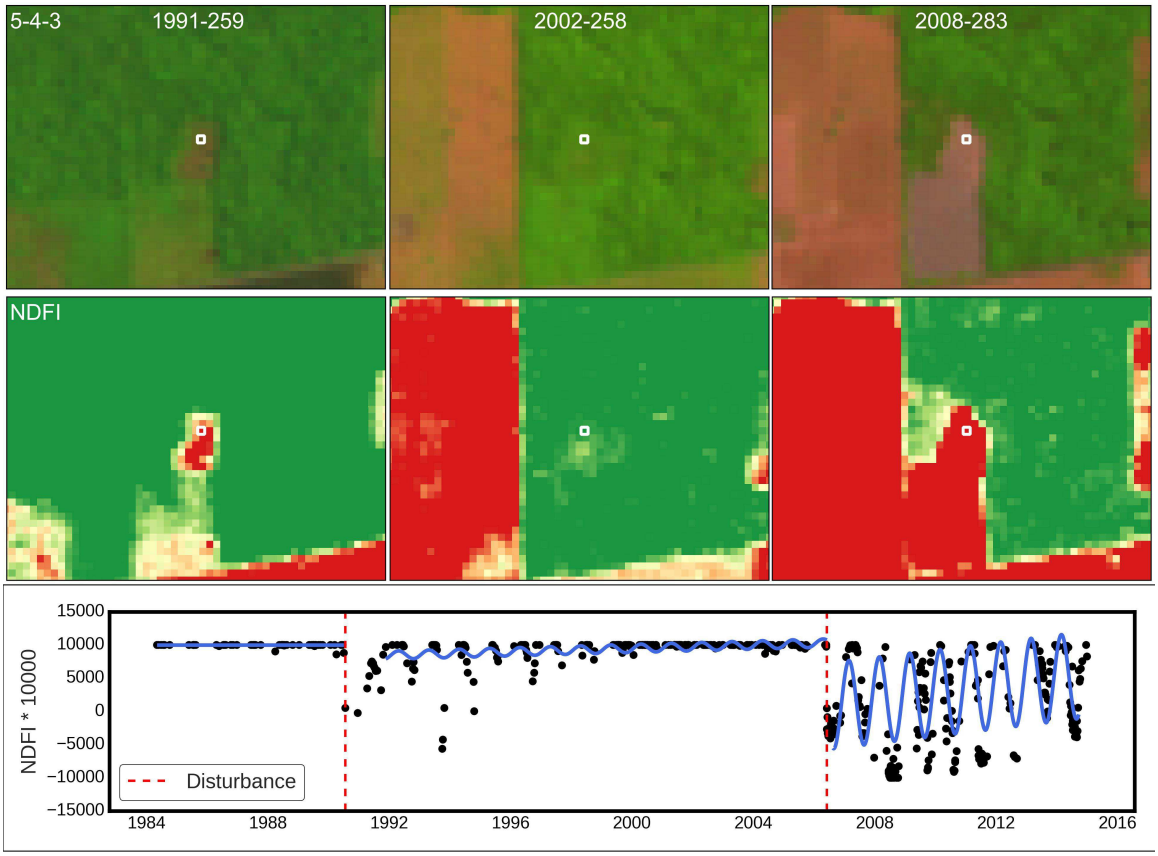


Figure 9. The time series for a location that undergoes two disturbances. The top row of imagery shows the original Landsat data in 5-4-3 stretch and different stages of the disturbance, the middle shows the calculated NDFI images, and the bottom shows the time series for the location showed in while in the images. A logging event causes a temporary change in the NDFI time series around 1990. The models on both sides of the disturbance are classified as forest and the disturbance is labeled as degradation. Later, a second disturbance causes a change in land cover. Notice the difference between low-magnitude and temporary degradation event and the higher magnitude deforestation event.

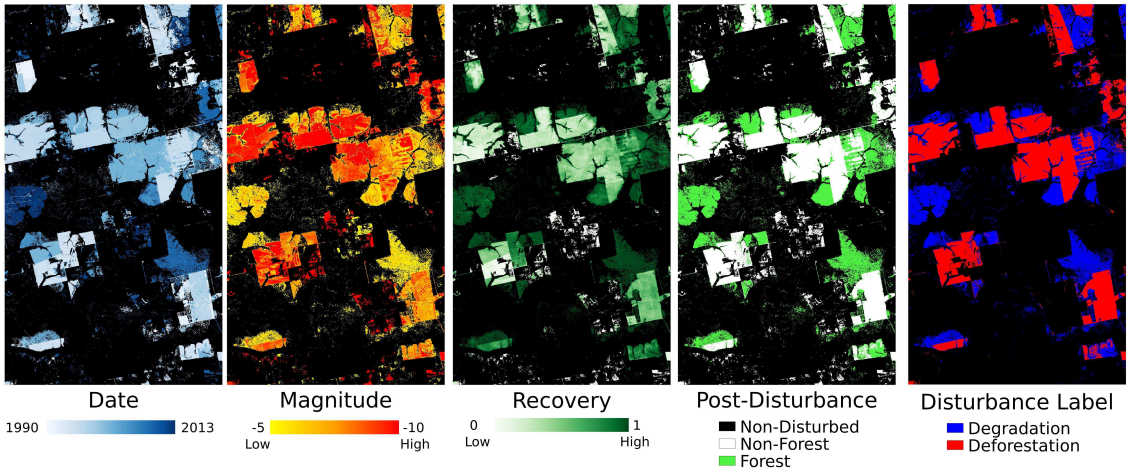


Figure 10. Output data calculated from the disturbance monitoring algorithm. The outputs include dates of disturbances, disturbance magnitude, post-disturbance recovery, post-disturbance land cover class, and disturbance label. The date, magnitude, and recovery are continuous numerical values that are simplified on the legend for visualization purposes.

For every disturbance detected by the algorithm there is an associated disturbance date, magnitude of change, level of recovery, classification of the pre- and post-disturbance land cover, and disturbance label (Figure 10). A spatial buffer and segmentation post-processing routine is applied to eliminate disturbance events that are spatially isolated and small in size and magnitude. Disturbance events in the region have been shown to be almost entirely near previous disturbances or development (Alves et al., 1998). Exceptions are natural disturbances such as fires or inundation, but these disturbances tend to result in a high change magnitude and therefore would not be removed during the post-processing. Therefore, we defined a set of criteria to remove unrealistic disturbance events. A disturbance is removed from the dataset if all of the following criteria were met:

1. The post-disturbance recovery is greater than .99, indicating that the intercept term for NDFI after the disturbance is at least 99% of the

magnitude of the intercept term before the disturbance. This criterion would exclude events that cause long-term changes in NDFI.

2. The event occurs in a location that is greater than 500 meters from a non-forest land cover or a previous disturbance event that does not meet these criteria.
3. There are no other disturbance events within a 1-pixel buffer and with a disturbance date within a 3-year window.
4. The change magnitude is greater than -6. Since the change threshold is -5, only very low magnitude changes would meet this requirement.

The post-processing routine removed spurious noise that was labeled as a disturbance in areas of stable forests. Because a location can contain multiple disturbances, it is possible that one of the disturbances is removed by the post-processing routine while others remain. While the post-processing routine can potentially remove isolated pixels, all four of the criteria must be met for it to be removed. Therefore, the resulting maps retain a minimum mapping unit (MMU) of 30m. Maps of deforestation and degradation can be created for any subset of time within the monitoring period (Figure 11).

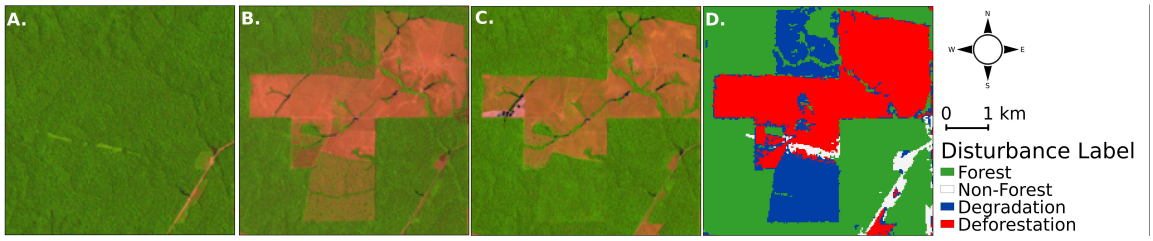


Figure 11. A. A Landsat 5-4-3 image before the study period on 1989-09-19, B. An image during the study period and in the middle of a logging event on 2005-10-17. C. An image after the study period on 2016-09-29. D. Disturbance classification for the period of 1990-2013. Notice the areas thinned in B. but appearing as forests in C. are classified as degradation while the areas converted to pasture are classified as deforestation.

Sample-based estimation of area and accuracy

Any map constructed by classification of satellite data will contain classification errors. Errors are inevitable as the translation of measurements of reflected sunlight into complex land surface processes – in particular forest degradation and post-disturbance recovery – is inherently difficult. If classification errors are present, the areas of land categories as depicted by the map will be incorrect. The estimation of areas is better accomplished by applying an unbiased estimator to a sample of reference observations (GFOI, 2016). The sample data are collected by probability sampling and the reference observations are the best available assessments of the reference conditions on the ground (Olofsson et al., 2014). In addition to adjustment for classification errors by the unbiased estimator, a sample-based approach to estimation allows for quantification of the uncertainty of area estimates by estimation of confidence interval. The sample data also allow for estimation of map accuracy by comparing map labels to the reference observations. Further, sample-based estimation of area is stipulated by the IPCC in their good practice guidelines for greenhouse gas inventories. Good practices state that inventories should be (IPCC, 2003) "neither over-nor underestimates so far as can be judged, and in which uncertainties are reduced as far as practical." These two criteria are directly related to bias, which is a property of an estimator, and uncertainty, which can be quantified by a confidence interval (GFOI, 2016). Therefore, land change information, obtained directly from a map through "pixel counting" is not consistent with proper reporting protocols. The rationale holds true also for forest degradation.

Because land change, and forest degradation in particular, tends to be a small part of the study area relative to areas of stable land categories, a stratification of the study area based on a map that depicts the categories of interest ensures sufficient statistical representation of rare categories such as deforestation and degradation (Olofsson et al., 2014). There are a variety of unbiased estimators available but when using discrete map classes to construct strata the stratified estimator has proven efficient (Cochran, 1977; GFOI, 2016; Olofsson et al., 2013). To test the efficiency of the maps of forest degradation and deforestation generated by the presented approach (and to assess their accuracy) in a sampling-based approach to estimation of the area of forest degradation in Rondônia, a map for the time period of 1990 to 2013, displayed in Figure 12, was produced to define strata (names of strata/map classes are italicized and capitalized):

- *Forest1*: Stable forest that did not exhibit any disturbances. A forest was defined as any pixel that contained over 10% tree cover and includes forested Cerrado.
- *Non-Forest*: Non-forested area including water, grassland, development, and agriculture. Land cover or condition changes can occur in the Non-Forest stratum as long as the land cover was not forest at the time of the disturbance.
- *Deforestation1*: Any area that was deforested within the study period that did not meet the requirements for the Deforestation2 stratum. Deforestation explicitly requires a conversion from forest to non-forest.

- *Deforestation2 (Possible Degradation)*: Locations that were classified as deforestation but were also identified as being potentially mis-classified as deforestation instead of degradation. The criteria for this stratum were applied to pixels that were classified as deforestation but had a post-disturbance recovery above .90 (Figure 10), indicating that the land cover after the disturbance corresponds to NDFI that is similar in magnitude to the observations before the disturbance. We hypothesize that pixels meeting this criterion have a high likelihood of containing errors of omission for degradation.
- *Degradation*: Any pixel that underwent at least one degradation event but no deforestation. Degradation explicitly requires a disturbance with no conversion of land cover.
- *Forest2 (Possible disturbance)*: A post-stratum, or a stratum created after the original stratification, for pixels classified as forest but identified as having a high-probability of having experienced a disturbance. The criteria for this post-stratum were a two-pixel buffer around disturbance events and any pixels that were originally labeled as a disturbance but removed due to the post-processing procedure.

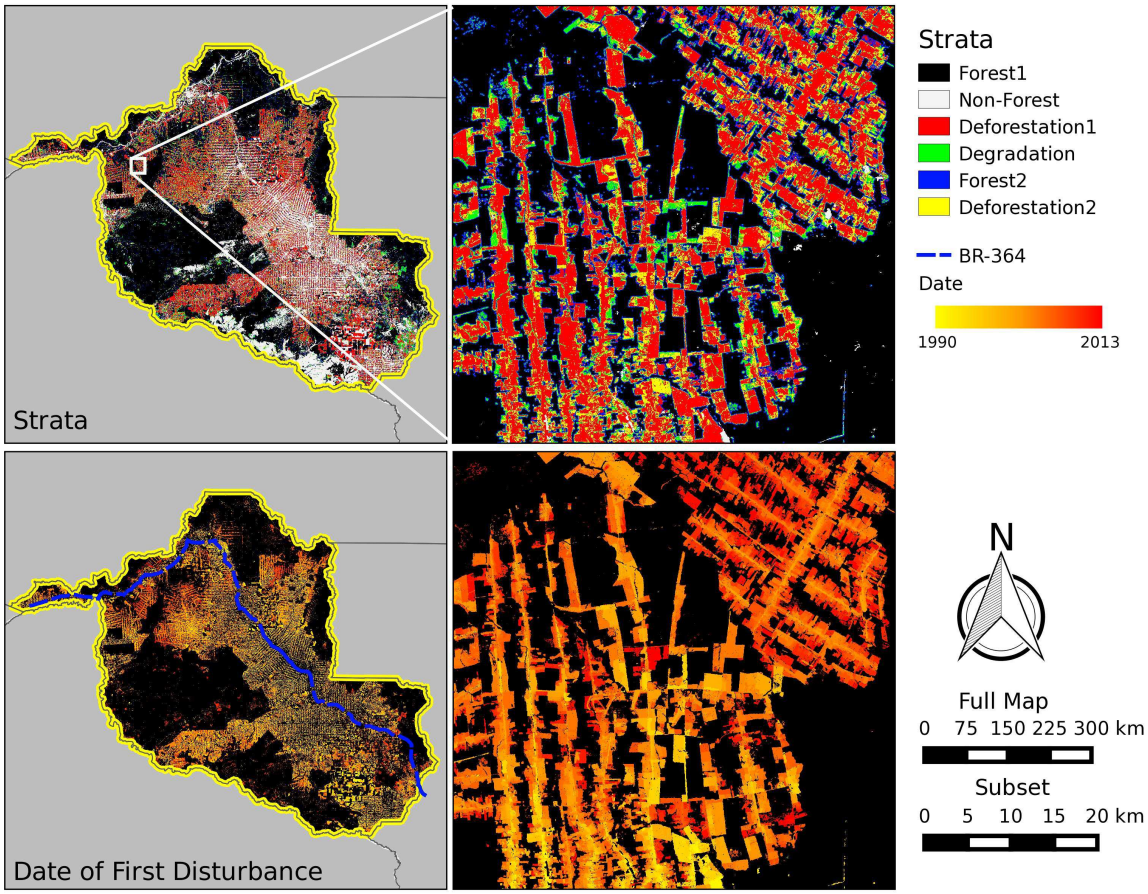


Figure 12. Map strata and date of first disturbance for forest disturbances from 1990 to 2013.

The *Deforestation2* and *Forest2* strata were included to improve the precision of the area estimate of *Degradation*. The impact of sample units in the large *Forest* stratum identified as degradation in the reference data – i.e. errors of omission of degradation – on the standard error of the area estimate of degradation is directly related to the area weight of the stratum that contained the omission (Cochran, 1977). Because the *Deforestation* stratum is much larger than the *Degradation* stratum, the goal of the *Deforestation2* stratum was to concentrate sample units identified as degradation but mis-classified as deforestation into a stratum with relatively small area weights. The *Forest2* post-stratum was used for the same

reasons, i.e. to concentrate omitted degradation and deforestation sample units in in a smaller stratum (*Forest1*). For a post-stratified estimator to be unbiased, however, the original sample needs to include pixels that are within the post-strata (Stehman, 2013). We hypothesized that the large sample size in *Forest1* (500; see next section) and the stratum weight of *Forest2* (.066) would ensure that sample units were within the post-stratum. Therefore, we used *Forest2* as a post-stratum and *Deforestation2* as part of the sampling design (i.e. a "pre-stratum"). The *Forest2* post-stratum is similar to the use of a spatial "buffer" stratum; an approach that becoming increasingly popular to contain the impact of omission error in forest monitoring applications (Arevalo et al., 2018; Hansen et al., 2016; Potapov et al., 2017; Tyukavina et al., 2013). The difference is that in our case, the *Forest2* stratum also incorporates information from the model to identify possible disturbance events instead of relying on spatial proximity to other disturbances.

Sampling and Response Design

Table 2. Strata area, weights, and sample allocation. Forest2 was not assigned any samples because it was a post-stratum.

	Forest1	***Non-Forest***	***Deforestation1***	***Deforestation2***	***Degradation***	***Forest2***
Area (ha)	10936185	4512933	4279446	1147116	1170166	1586974
Wi (%)	46.26	19.12	18.12	4.845	4.99	6.66
Samples	500	175	175	75	75	0

We calculated the total sample size for our stratified random sample using Equation 13 in Olofsson et al. (2014) but modified to set a target standard error of the area of

535 degradation as opposed to overall map accuracy. Targeting a 10% margin of error,
536 defined as two standard errors divided by the area estimate of degradation yielded
537 a sample size of 1,000 sample units (because the area of degradation was unknown
538 at the sampling stage, we used the area of the *Degradation* stratum as a proxy when
539 determining the sample size). The sample was allocated to strata according to "Good
540 Practices" as defined in Olofsson et al. (2014) (Table 2) for targeting area estimation
541 of degradation. The selection of training data was completely independent from the
542 design and selection of the reference sample.

543 Forest degradation and post-disturbance recovery are complex processes and
544 collecting reference observations of degradation and recovery is an inherently
545 complicated and difficult task. Relying on high-resolution data is insufficient as it
546 does not allow for a reconstruction of landscape dynamics; instead, after
547 investigating a set of degradation and recovery events, it was determined that time
548 series of Landsat observations with the support of high-resolution data were
549 required. While this combination of data makes for a powerful reference dataset, the
550 amount of data required for each sample location is considerable.

551 Therefore, to facilitate collection of reference observations, an application was
552 developed in Google Earth Engine that allowed for visualization of time series of all
553 Landsat data ever acquired for the locations of the sample units, without having to
554 download the data (see github.com/bullocke/coded for access to the interpretation
555 tool, reference sample, and algorithm code). We used the historical time series of
556 Landsat in addition to high-resolution imagery on the GEE web interface to assign
557 reference labels to the sample units. Four trained interpreters analyzed the sample.

The interpreters did not have knowledge of the strata to which the sample units were allocated when assigning reference labels. All disturbances in the historical time series of each sample unit were labeled as either degradation or deforestation. The interpreter's confidence in the reference label was also recorded, and any label with low or medium confidence was examined by at least two interpreters. All sample units labeled as disturbance were also analyzed by at least two interpreters. We ultimately needed to remove four sample units, for which it was impossible to assign a reference label, making a total of 996 sample units.

Results

Accuracy Assessment and Area Estimation

The results of the accuracy assessment and area estimation are in Table 3. We estimate that 53% ±1.13% (95% confidence interval) of the State of Rondônia remained undisturbed forest through 2013, with 6% ±0.89% of the forest having undergone degradation, 16% ±1.47% being non-forest, and 25% ±1.32% being deforested. In total, we estimated that 7,464,140 ±318,979 hectares of forest was disturbed (combination of deforestation and degradation). The outcome of the accuracy assessment indicates that the map under-represented both degradation and deforestation by -30.1%, -9.5%, and over-represented non-forest by +18.5%. The margins of errors of the area estimates (defined as the ratio of the half width of the 95% confidence intervals to the estimate) were 13.9% and 5.3% for degradation and deforestation. If we had not post-stratified and used only one forest stratum

579 (*Forest1* + *Forest2*), and assuming we only sampled one deforestation stratum, the
580 margin of error of the degradation area estimate would increase from 14% to 17%,
581 which is equivalent to a 10% increase in the standard error of the area estimate. The
582 User's and Producer's accuracy for *Degradation* was 88.0% and 68.1%, while for
583 *Deforestation* (i.e. *Deforestation1* + *Deforestation2*) it was 93.3% and 85.3%. The
584 overall accuracy of the map was 90.8%. Without the buffer strata, the User's and
585 Producer's accuracies for degradation would decrease to 78.7% and 63.4%
586 accordingly, and Overall accuracy would decrease from 91.0% to 88.8%.
587

588 **Table 3. Confusion matrix shown as sample counts and proportions of area estimates,**
589 **area estimates of each category derived from the stratified random sample, and**
590 **accuracy measures. The first (top) confusion matrix of sample counts compares the**
591 **reference data to the stratification, while the second one compares the reference data**
592 **to the stratification if it were made without the buffer strata. Note that the second**
593 **matrix is for reference purposes only, and all estimation was performed using the**
594 **first matrix.**

	Reference Data			
	Forest	**Non-Forest**	**Deforestation**	**Degradation**
Map Strata	**Sample Counts**			
Forest1	418	8	1	0
Non-Forest	12	128	29	5
Deforestation1	0	4	169	3
Degradation	6	1	2	66
Forest2	64	1	3	5
Deforestation2	0	1	60	11
Map Class	**Sample Counts (No Buffer)**			
Forest	482	9	4	5
Non-Forest	12	128	29	5

Deforestation	0	5	229	14
Degradation	6	1	2	66
	Area Proportions			
Forest1	0.453	0.009	0.001	0.000
Non-Forest	0.013	0.141	0.032	0.005
Deforestation1	0.000	0.004	0.174	0.003
Degradation	0.004	< 0.001	0.001	0.044
Forest2	0.058	< 0.001	.003	0.005
Deforestation2	0.000	< 0.001	0.040	0.007
Area	.528	.156	.251	.064
Area [ha]	12489235	3679448	5941628	1522512
95% CI [ha]	266342	347341	312746	211266
Margin of Error [%]	2.13	9.44	5.26	13.88
Map Bias [%]	+0.27	+18.5	-9.49	-30.1
User's Acc. [%]	96.6	73.6	93.3	88.0
Producer's Acc. [%]	96.8	90.3	85.3	68.1
Overall Accuracy [%]	91.02%			

595

596 *Disturbance rates*

597 Yearly mapped areas of disturbances (both deforestation and degradation) that
598 were detected by our algorithm can be seen in Figure 13. The areas represent the
599 sum of all the degradation or deforestation events for each year. If a pixel undergoes
600 multiple disturbance events it would therefore be included multiple times in the
601 annual calculation of mapped area. Deforestation mostly increased throughout the
602 1990s and peaked between 1996 and 2004 and then dropped sharply after 2004,

during the period that the Brazilian government was beginning to implement PPCDAm (PPCDam, 2004). Degradation rates rose during the 1990s but remained high until 2010 (Figure 13). Interestingly, the degradation peaked immediately after the implementation of PPCDAm in the period from 2005 to 2010. Deforestation rates were the lowest from 2007 through 2013. Degradation rates were lowest at the beginning of the time series from 1990 through 1994.

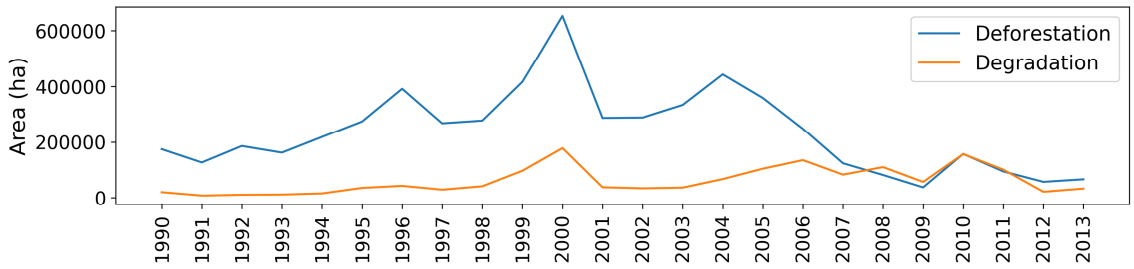


Figure 13. Yearly area of deforestation and degradation. Note that deforestation generally trended downward after 2004 while degradation remained high until 2010. Both degradation and deforestation were highest in 2000.

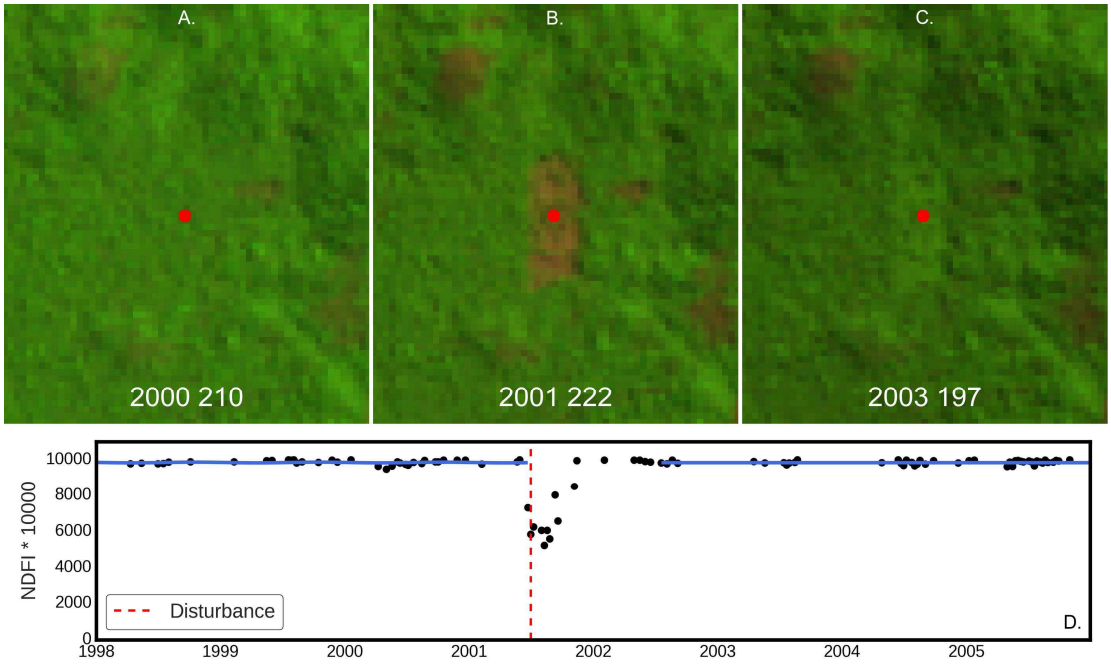
Discussion and Conclusions

Despite an increase in the quality and quantity of satellite data, processing power and investments in forest monitoring systems over the last decade, forest degradation remains a big unknown in many assessments of the impact of human activities in tropical forests. Effective systems for monitoring forest degradation over large areas are required to properly assess and estimate such impacts. By utilizing freely available Landsat data on the Google Earth Engine, we have developed a methodology for large-scale monitoring of forest degradation and deforestation. The approach was shown to be accurate, with a 91% overall accuracy of a map of forest degradation and deforestation in Rondônia, Brazil. The map

623 products efficiently stratified the study area to allow for unbiased estimation of the
624 areas of deforestation and forest degradation; margins of error of 5% and 14%
625 respectively at the 95% confidence level were achieved.

626 There are three components of our approach that are essential for the detection
627 of disturbances and, in particular, degradation: characterizing the pre-disturbance
628 or "stable" signal in the data, distinguishing between disturbances and regular
629 variability, and determining the post-disturbance land cover. For all of these
630 components it is advantageous, if not essential, to utilize repeated and historical
631 data in the analysis. The ability to characterize the "natural" state of the forest is
632 problematic when using an approach that relies on a single high-to-medium
633 resolution image. A thinned forest may, for a limited amount of time, appear
634 structurally similar to a forest with a naturally open canopy. However, if the pattern
635 is consistent over multiple years then it is likely that it represents the stable state of
636 the undisturbed forest. Our approach characterizes the forest during the training
637 period by fitting regression models to the time series of the data. The intercept,
638 seasonal harmonics, and RMSE of the models represent the average magnitude,
639 intra-annual, and random variability in the data. An area that has been thinned may
640 look similar to an open canopy in a single image, but the pattern over the 6-year
641 training period will be different. Therefore, our approach is able to differentiate
642 between canopy gaps due to disturbance and forests with naturally open canopies
643 and thus can be used for monitoring in forests with varying crown cover (see Figure
644 4,Figure 7).

645 The spectral response of a degradation event in optical remote sensing data can
646 be minimal and therefore, to accurately identify a disturbance, the change must be
647 effectively differentiated from ephemeral "noise" in the data. If a single observation
648 exceeds the average variability of the regression model (or the model's RMSE), it
649 can be assumed to be "noise" and discarded from analysis. However, if there are
650 consistent observations that are beyond the natural variability then it is likely that
651 there has been a change in the forest. Therefore, the use of many observations
652 affords the opportunity to separate disturbances from the inherent variability in the
653 observations, or the "noise." Since the time series of NDFI in Rondônia tend to
654 contain minimal natural variability (Figure 5), a disturbance in the forest does not
655 need to be large for a degradation event to be identified. The ability to characterize
656 the background variability in undisturbed forests using many Landsat observations
657 allows for the detection of degradation events that cause minimal spectral changes
658 in the data. Additionally, the impact from the disturbance event does not need to
659 prevalent for a long time period when using continuous data. An example of a
660 disturbance can be seen in in Figure 14 in which a logging event in a forest results in
661 a change in NDFI that is only prevalent for a few months. A technique based on a
662 single image outside the period in which the damage is noticeable would miss the
663 disturbance entirely.



664

665 **Figure 14. Landsat 5-4-3 images before (A.), during (B.), and after a logging event (C.),**
666 **and the NDFI time series of a location shown in red (D.). Notice that the disturbance is**
667 **only prevalent in the NDFI time series for a few months.**

668

669 The definitions of deforestation and degradation are directly related to the land
670 cover after the disturbance and, as a result, they cannot be effectively differentiated
671 without a characterization of the post-disturbance landscape. By fitting new
672 regression models to the data after the disturbance our approach can determine
673 whether the event resulted in a change in land cover and, as consequence, whether
674 the disturbance was deforestation or degradation. Since this process is run
675 continuously through time, maps of forest disturbances can then be derived for any
676 temporal subset within a study period. However, for very recent disturbance events
677 it is difficult to differentiate deforestation and degradation, as observations after the
678 disturbance are necessary. These factors demonstrate that time series analysis is
679 highly valuable for the accurate monitoring of degradation.

680 The use of the *Forest2* post-stratum and *Deforestation2* pre-stratum improved
681 the precision of the area estimate of degradation by decreasing the area weight of
682 units of omitted degradation in the sample data. The impact of omissions of forest
683 disturbance in areas mapped as stable forest have been documented in other
684 studies, and methods to minimize the effect of such errors have been suggested
685 (Olofsson et al., 2018). In essence, the often critical impact of omissions errors is the
686 result of the weights of the strata in which the omission errors occur. The objective
687 of the *Forest2* and *Deforestation2* strata was to reduce the area of the strata (i.e. the
688 strata weights) that contained the omissions, such that the larger *Forest1* and
689 *Deforestation1* strata would be free of omissions. All five errors of omission of forest
690 degradation in the sample data -- i.e. the sample units located in*Forest1* but labeled
691 degradation in the reference data -- ended up in the *Forest2* post-stratum. The result
692 was a reduction of the area weight of the omission errors. Additionally, 11 of the 14
693 sample units of degradation in the *Deforestation1* stratum ended up in the
694 *Deforestation2* stratum. Again, this reduced the weight of these 11 omission errors
695 of degradation because of the smaller size of *Deforestation2* compared to
696 *Deforestation1*. The approaches used in this study to minimize the impact of
697 omission errors were based on buffering of the change strata in addition to model-
698 informed information on the areas of high-likelihood change, such as disturbance
699 magnitude and recovery. Based on our results, we recommend the use of model-
700 informed information to generate strata with a high likelihood of containing errors
701 of omission for the target stratum. No testing was performed on the optimal size of
702 these strata and we suggest this as an area for future research.

703 The Producer's Accuracy of *Degradation* was relatively low (68.1%) compared to
704 the other classes due to errors of omission, or degradation falsely labeled as non-
705 disturbed or deforested. Most of the omissions of degradation were misclassified as
706 deforestation: 14 out 24 sample units identified as degradation in the reference data
707 occurred in one of the two deforestation strata (Table 3). The result suggests that
708 for most of the omissions of degradation, the algorithm correctly detected the
709 disturbance but did not identify the correct post-disturbance land cover by labeling
710 the disturbance *deforestation* instead of *degradation*. Additionally, five of the
711 omissions of degradation occurred in the *Non-forest* stratum and, because the
712 change detection procedure is not applied to time series models classified as *Non-*
713 *forest* , these five omission errors were also influenced by forest/non-forest
714 misclassification (Figure 3). The omission errors of degradation and deforestation in
715 the *Non-forest* stratum would be reduced or eliminated by simply improving the
716 forest and non-forest classification. Improved differencing between the post-
717 disturbance land covers will reduce the amount of degradation mis-classified as
718 deforestation, and by incorporating the post-disturbance recovery into the
719 classification procedure. The post-disturbance recovery was used as the criteria for
720 creating the *Deforestation2* stratum, which contained 11 of the 14 samples mapped
721 as deforestation but were degradation in the reference data.
722 In contrast to errors due to mis-classification, five of the sample units (or 0.5%
723 of the sample) that were identified as omissions of degradation occurred in the
724 *Forest2 stratum* class and were therefore caused by errors in the change detection.
725 In most of these cases, we found that the omitted disturbance caused a change in

NDFI that was either too low of magnitude or too short in duration to pass our change threshold and/or consecutive observation requirements. The change thresholds were developed after extensive testing and chosen to minimize errors of commission. While reducing the change thresholds would reduce the likelihood of omission errors, it would increase the likelihood of commission errors of disturbance. Regardless, judging from the accuracy of mapped degradation and the standard error of estimated degradation, we believe an omission rate of 1 in 100 (or 5 of 500 sample units) for degradation mapped as *Forest* to be acceptable given the historical difficulties in detecting degradation (Herold et al., 2011).

It is possible that the consecutive observation threshold will cause a higher omission rate in areas with low data availability. In cloudy regions of the tropics there may not be five clear observations for an entire year and, therefore, changes will be omitted that are only visible for a short amount of time (Figure 14). This limitation may be addressed by incorporating data from other sensors such as Sentinel-2. Alternatively, a consecutive observation threshold that is dependent on data availability may be able to reduce errors of omission due to data gaps. We recommend this as an area for future research.

In the study area from 1990 to 2013, we estimated that the extent of forest degradation was 24% of the area deforested. It is possible, and in many cases likely, for locations to undergo multiple degradation events or be degraded before deforested (Figure 9). Of all the locations that underwent a disturbance, 33% were found to have been disturbed more than once. Therefore, the total amount of degradation is likely to be higher than our area estimate suggests. The degradation

area also includes different types of disturbance and each type will have different ecological impacts and resulting emissions. For example, the carbon emissions associated with fires will be different from that of selective logging or deforestation. The intensity of the fire or logging activity will also influence the associated emissions. We believe that the *change magnitude* calculated from our model is an indicator of the degree of canopy damage from a disturbance event (Figure 15). Comparing the change magnitude to direct damage on the ground is an area for future research.

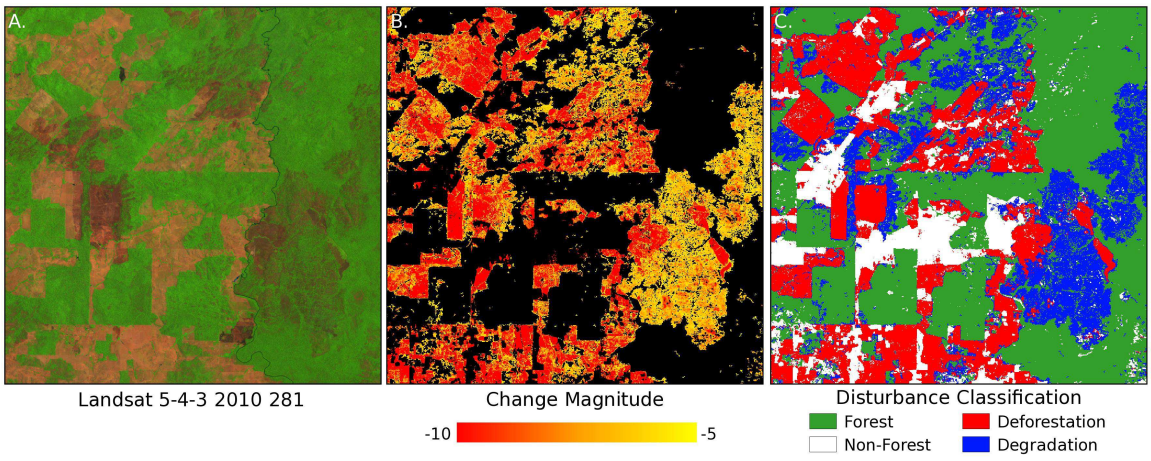


Figure 15. A. Landsat image from Path/Row 230/68 in Southeastern Rondônia showing areas that have been deforested and also burn scars. B. The change magnitude associated with disturbance events detected by the model. C. Land cover and disturbance classification. Note that the fires result in a lower change magnitude than the land cover conversions and also are classified as degradation.

Over the 24-year time period of our study there were clear temporal patterns in deforestation and degradation. The yearly rates of both types of disturbance rose rapidly in the late 1990s and early 2000s. After 2004, however, deforestation dropped rapidly and degradation rose to its highest level in the study period (Figure 16). We offer two possible explanations for this increase in degradation. First, PPCDAm included policies that dis-incentivize deforestation and incentivize

769 conservation (PPCDAm, 2004). Therefore, it is possible that some of the logging that
770 had previously occurred during the transition to alternative land covers is now
771 being followed by forest regrowth and would therefore be considered degradation.
772 Secondly, forest fires in the Amazon have been shown to be increasing in intensity,
773 with 2010 having an especially large number of fires (Aragão et al., 2018; Gatti et al.,
774 2014). This hypothesis is supported by the fact that our results show 2010 to be a
775 year with lots of degradation (Figure 13). This result is important because it suggests
776 that the increase in environmental services and the carbon emissions reductions
777 associated with decreased deforestation may be partially offset by forest
778 degradation. However, given the limited geographic extent of our research and lack
779 of disturbance type attribution, no conclusions can be made on the drivers of these
780 trends. National studies are needed to fully understand the interaction of
781 deforestation and forest degradation in Brazil.

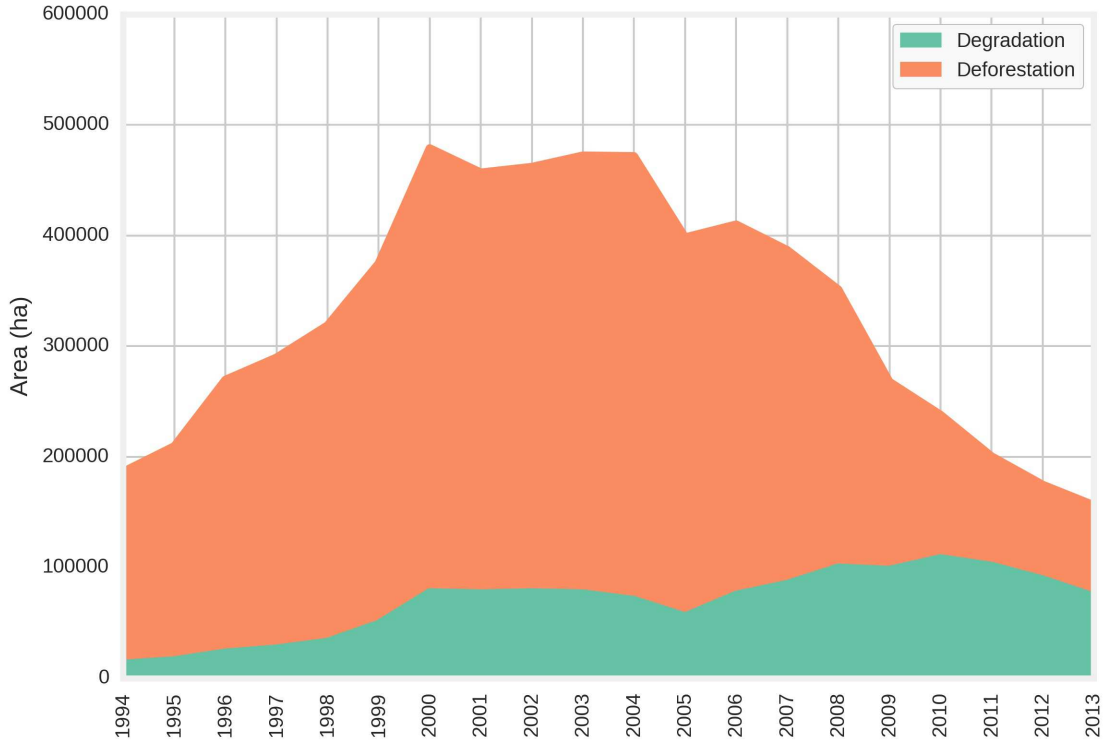


Figure 16. 5-year mean disturbance areas for Rondônia. Note that degradation mostly increased over the entire study period, while deforestation peaked in the early 2000s and then sharply decreased.

Finally, we did not attempt to quantify afforestation or secondary forests. We restricted the disturbance detection to models that were classified as forests during a training period to focus on the objective of detecting degradation. The change detection could easily be broadened to track changes in many types of land covers, as has been the primary focus of the CCDC algorithm, which combined with NFDI (Souza et al., 2005), was the motivating algorithm behind our approach. Arevalo et al. (2018) utilized the CCDC algorithm to track conversion between a variety of land categories in Colombia. Estimation of the areas of multiple types of land conversions, including gain of secondary forest, was performed on a biyearly basis. While Arevalo et al. (2018) demonstrated how IPCC-compliant estimation of land

conversion can be achieved using time series analysis of Landsat data, they did not map or estimate degradation. A primary focus in the future will be incorporating our research into an IPCC-complaint approach that tracks land conversion, degradation, and afforestation or forest regrowth.

Our methodology has shown that historical Landsat data can effectively be used to monitor and estimate forest degradation. The research demonstrates that the Landsat archive is an invaluable resource for land change monitoring. Historical data can be used to develop a "baseline" of the natural condition of a landscape. Subtle changes can be distinguished from ephemeral variations in the data by comparing repeated observations to this "baseline." The characterization of post-disturbance dynamics can then be used to differentiate degradation from deforestation. We hypothesize that such approaches will become essential for achieving a comprehensive analysis of ecosystem dynamics including forest degradation and other subtle changes in a landscape.

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